

ASSIGNMENT - I

apsara

Date:

PART - A

1. Differentiate between absolute and secondary instruments with examples.

Ans →

ABSOLUTE INSTRUMENTS

Absolute instruments are those instruments which indicate the quantity to be measured in the term of constant of instruments (dimension, turns etc). & in order to find out the quantity in the practical units it is necessary to multiply such deflection with an instrument constant.

No previous calibration or operation is necessary in this case.

The most common absolute instrument is tangent galvanometer.

SECONDARY INSTRUMENTS

Secondary instruments are those instruments which gives the magnitude of electrical quantity to be measured directly.

These instruments are required to be calibrated by comparing with a standard instrument.

These instrument are generally used in practice. Eg: Ammeter, Voltmeter, Wattmeter, pressure gauges etc.

2. What is the function of a damping system in indicating instruments?

Ans → The damping system in indicating instruments prevents oscillations of the pointer and ensures it quickly settles at the correct reading. When the measurement is taken, the pointer tends to overshoot due to inertia and may oscillate around the final reading before settling. Damping applies a resisting force to smooth this motion, improving accuracy and readability. Common types include air friction, fluid friction, & eddy current damping. Overall, it ensures fast, stable, and precise pointer movement without delays or fluctuations.

3. State the advantages and limitations of PMMC instruments.

Ans → Advantages:-→

1. The PMMC instrument have high accuracy because of the high torque weight ratio.
2. The PMMC instrument use shelf shielding magnet which is useful for the aerospace application.
3. Power consumption of these devices are very less.
4. No hysteresis loss, ensuring reliable readings.

Limitations →

1. The PMMC instruments are only used for the direct current.
2. Expensive due to precision construction.
4. What is the purpose of control springs in PMMC instruments?

Ans → Control springs in PMMC instruments provide the restoring torque that balances the deflecting torque, allowing the pointer to settle accurately. They also act as electrical conductors to the moving coil. Additionally, they return the pointer to zero when no current flows, ensuring precise and stable readings.

5. List the dynamic characteristics of measuring instruments.

Ans → Dynamic characteristics define how an instrument responds to changing inputs. Key characteristics are:

1. Response Time - Time taken to reach steady output after input change.
2. Dynamic Error - Difference between true and measured value during transients.

3. Lag - Delay between input change and output response.

6. Compare moving iron (MI) and moving coil (PMMC) instruments.

Any →	FEATURE	MOVING IRON (MI)	MOVING COIL (PMMC)
1.	Type of current	Measures both AC and DC.	Measures only DC.
2.	Scale	Non - Linear	Linear
3.	Accuracy	Moderate	High
4.	Sensitivity	Lower	Higher

7. Define sensitivity and resolution in measuring systems.

Any → Sensitivity :- The ratio of change in output signal to the change in input signal. If the sensitivity is constant then the system is said to be linear system. If the sensitivity is variable then the system is said to be non-linear system. It shows how responsive an instrument is. Higher sensitivity means the instrument can detect small input changes.

$$\text{Sensitivity} = \frac{\text{Change in output}}{\text{change in input}} = \frac{\Delta q_o}{\Delta q_i}$$

Eg: A voltmeter deflecting 1mm for every 1mV has 1mm/mV sensitivity.

Resolution $\rightarrow$ The smallest increment in input or input change which can be detected by an instrument is called as resolution or discrimination. It defines the instrument's ability to distinguish fine differences.

Eg: A thermometer with 0.1°C resolution can detect temperature changes as small as 0.1°C .

8. Explain the working of an electrodynamometer type voltmeter.

Ans $\rightarrow$ An electrodynamometer-type voltmeter works on the principle of the interaction between magnetic fields produced by two coils: a fixed coil & a moving coil. When a voltage is applied, current flows through both coils. The fixed coil produces a magnetic field, and the moving coil (attached to a pointer & spring) also generates a field due to the same current.

This interaction produces a deflecting torque on the moving coil, causing the pointer to move. The deflection is proportional to the square of the current and thus to the voltage. It can measure both AC and DC and uses spring control and air friction damping.

9. what are the basic parts of an indicating instrument?

Ans → Basic Parts of an indicating instrument are →

1. Deflecting System - Produces a force proportional to the input (eg - current or voltage) to move the pointer.
 2. Controlling System - Provides a restoring force (usually springs) to balance deflection and return the pointer to zero.
 3. Damping System - Prevents oscillations and helps the pointer settle quickly.
 4. Pointer and scale - Displays the measured value.
10. Define transducer. Give two examples & classify them.

Ans → It is a device that convert one form of energy into another form of energy, typically converting a physical quantity into an electrical signal.

- Eg:
1. Thermocouple converts heat to voltage.
 2. LVDT converts displacement to voltage.
 3. Strain gauge converts mechanical strain into resistance change.

Transducers can be classified as:

1. Active → Generates output without external power. Eg - Piezoelectric sensor.
2. Passive → Requires external power to operate. Eg - Strain gauge.

PART - B

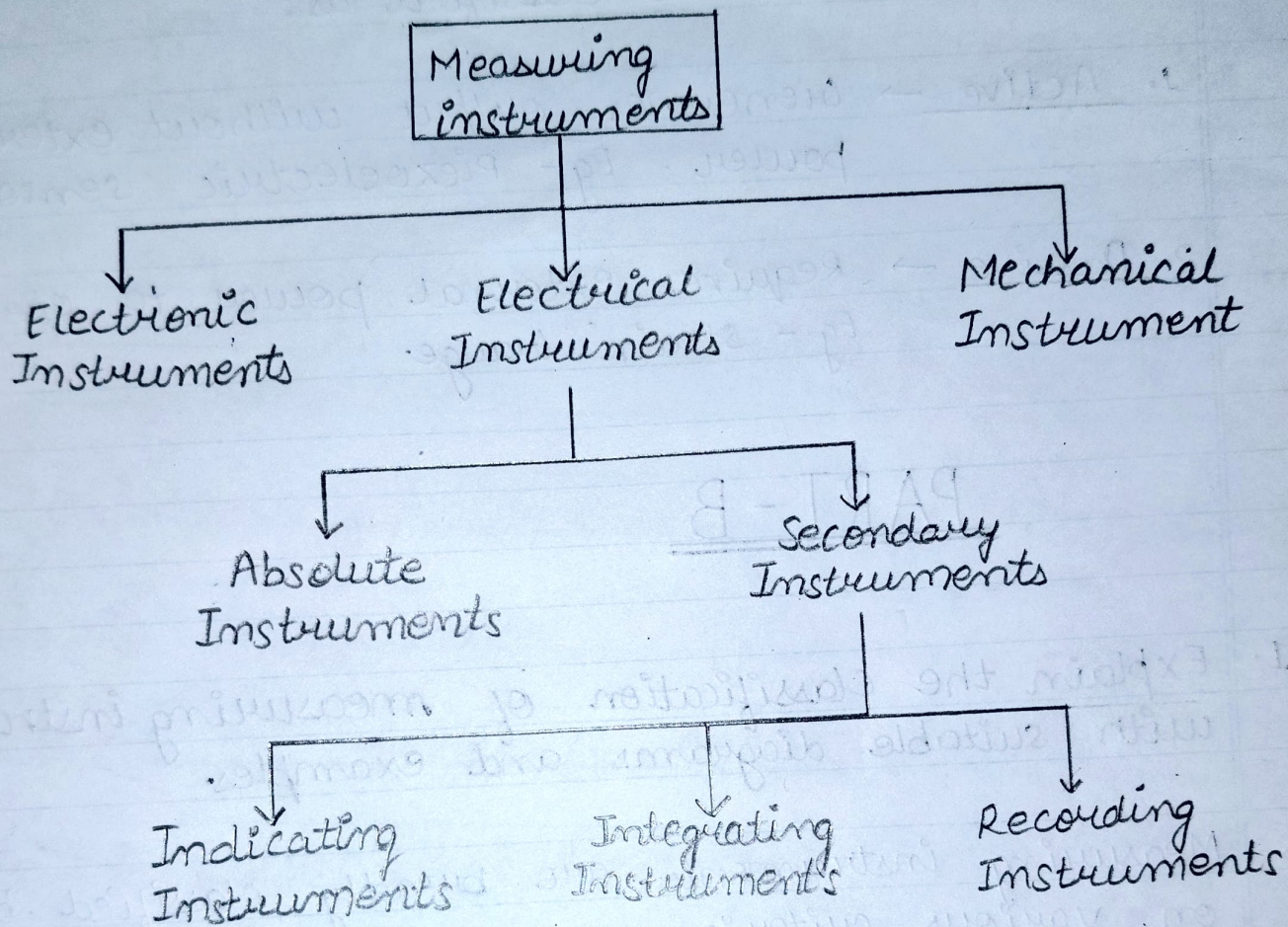
1. Explain the classification of measuring instrument with suitable diagrams and examples.

Ans → Measuring instruments are broadly classified based on various criteria.

1. Based on Function →

(a) Absolute Instruments → Measure quantity using physical laws without prior calibration. Ex → Tangent galvanometer.

(b) Secondary Instruments → Require calibration against standards; indicate or record measured quantity. Ex → Voltmeter, Ammeter.



2. Based on output Type:- →

(a) Indicating instruments → show reading through a pointer on a scale.

Eg: Analog voltmeter.

(b) Recording Instruments → Record values over time on paper or digitally.

Eg: ECG machine.

(c) Integrating Instruments → Measure total quantity over a period.

Eg: Energy meter.

3. Based on Principle of operation →

(a) Electromechanical Instruments → Use mechanical movement & electrical inputs. Eg: PMMC instrument.

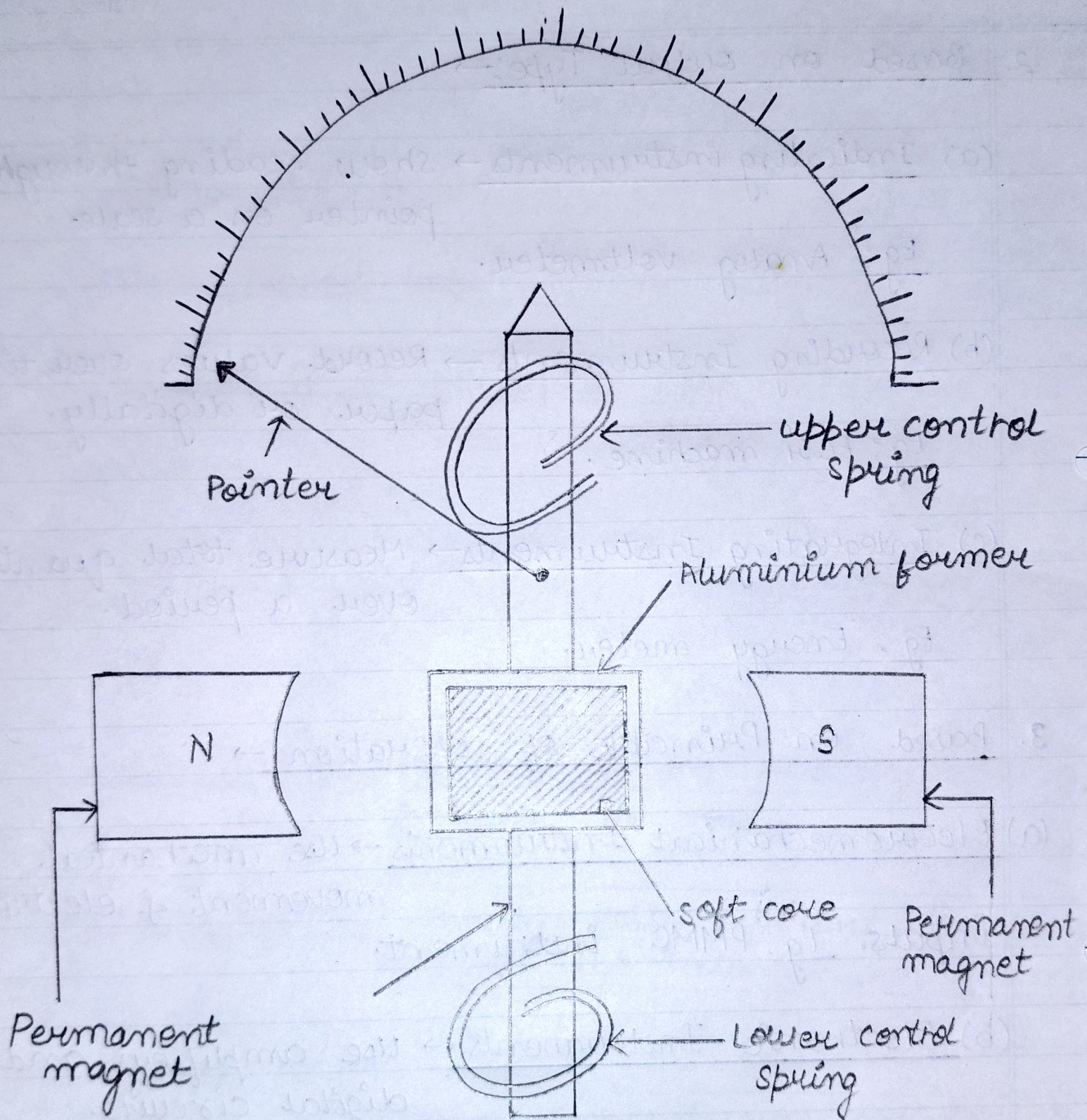
(b) Electronic Instruments → Use amplifiers and digital circuits.

Eg: Digital multimeter.

2. With the help of a neat sketch, explain the construction and working of a PMMC instrument.

→ Construction →

A PMMC instrument consists of the following main parts: →



INSTRUMENT FOR PMMC

- Permanent Magnet : Provides a strong and constant magnetic field.
- Moving Coil : A rectangular coil of fine wire wound on a light aluminium frame, placed between the magnetic poles.
- Soft Iron Core : Cylindrical core inside the coil to make the magnetic field radial and uniform.
- Control springs : Provide the controlling torque and also carry current to the coil.
- Pointer & Scale : Attached to the moving coil to indicate the reading.
- Damping System : Usually eddy current damping using the aluminium frame.

Working →

When a DC current flows through the coil, it experiences a torque due to the interaction between the magnetic field of the permanent magnet and the current in the coil (Lorentz force). This torque causes the coil to rotate. The controlling torque from the springs balances the deflecting torque, & the pointer settles at a position proportional to the current. Since the scale

is linear, the reading is accurate and easy.

3. Discuss the constructional features and working of an MI instrument. why is it suitable for both AC and DC?

Ans → Construction →

- Fixed Coil: Produces a magnetic field when current flows.
- Moving Iron: Soft iron piece (attraction or repulsion type) moves in response to the field.
- Pointer & Scale: Shows the reading.
- Control springs: Provide restoring torque.
- Damping: Usually air friction.

Working →

Current through the coil generates a magnetic field. The moving iron is attracted or repelled, causing the pointer to deflect. Deflection is proportional to the square of the current, so it's always in one direction.

Since, deflection depends on the square of the current, direction doesn't matter. This makes MI instruments suitable for both AC and DC measurements.

4. Explain the working principle and applications of electrodynamometer - type wattmeter.

Ans → Working principle →

It operates on the interaction between the magnetic fields of a fixed current coil and a moving voltage coil. The deflecting torque is proportional to $V \times I \times \cos(\phi)$ measuring true power in AC and DC circuits.

Applications :

- Measures AC and DC power.
- used in laboratories and calibration setups.
- Ideal for precision power measurement tasks.

5. Compare PMMC, MI and Electrodynamometer instruments on the basis of construction, operation and application.

Ans →	FEATURE	PMMC	MI	ELECTRODYNAMOMETER
	construction	Moving coil in Permanent magnetic field; linear scale.	Soft iron piece inside coil; simple and robust design.	Fixed and moving coils; no permanent magnet.
	operation	Deflecting torque due to current in	Deflection due to magnetic attraction/repulsion	Torque from interaction of fields in fixed

FEATURE	PMMC	MI	ELECTRODYNAMOMETER
	coil in magnetic field (Lorentz force); works on DC only.	of iron; works on AC and DC.	and moving coils; works on AC and DC.
Applications	used for DC ammeters, Voltmeters	used for AC/DC voltmeters & ammeters.	used in precision AC/DC wattmeters & power measurement.

6. Describe in detail the static and dynamic characteristics of measuring instruments with proper definitions and examples.

Ans → Static characteristics → These refer to the performance of an instrument when measuring a constant (non-varying) input.

(a) Accuracy → Closeness of the measured value to the true value.

Eg: A voltmeter showing 9.98 V for an actual 10 V is highly accurate.

(b) Precision → Ability to reproduce the same measurement under unchanged conditions.

Eg: A scale that gives the same weight reading every time for the same object.

(c) Sensitivity → Ratio of change in output to change in input.

Eg: A thermometer showing 0.1°C rise for a 0.1°C change is highly sensitive.

(d) Resolution $\rightarrow$ Smallest measurable input change that can be detected.

Eg: A digital voltmeter with a resolution of 0.01V .

Dynamic characteristics $\rightarrow$ These describe how an instrument responds to time-varying inputs.

(a) Fidelity $\rightarrow$ Ability to reproduce input variations accurately without distortion.

Eg: A sound level meter showing the exact change in noise level.

(b) Lag $\rightarrow$ Time delay between input change and output response.

Eg: A temperature sensor taking time to register a sudden temperature change.

(c) Dynamic error $\rightarrow$ The difference b/w true value & instrument reading during transients.

Eg: An ammeter may overshoot when current suddenly increases.

(d) Response Time $\rightarrow$ Time taken by the instrument to reach a steady value after input is applied.

Eg: A thermometer taking 5 seconds to show the final temperature.

7. Explain the construction and working of an induction-type wattmeter. Where is it commonly used?

Ans → Construction →

An induction-type wattmeter consists of:

- Shunt Coil (Voltage coil): Connected across the load; produces a magnetic field proportional to voltage.
- Series coil (Current coil): Connected in series with the load, produces a field proportional to current.
- Aluminium Disc: Placed between the two coils; free to rotate.
- Permanent Magnet: Provides damping torque to stop oscillations.
- Pointer and scale: Attached to indicate power reading.

Working →

The voltage and current coils produce alternating magnetic fields that induce eddy currents in the aluminium disc. The interaction of these eddy currents with the magnetic fields generates a torque that causes the disc to rotate. The speed of rotation is proportional to

true power ($V \times I \times \cos \phi$).

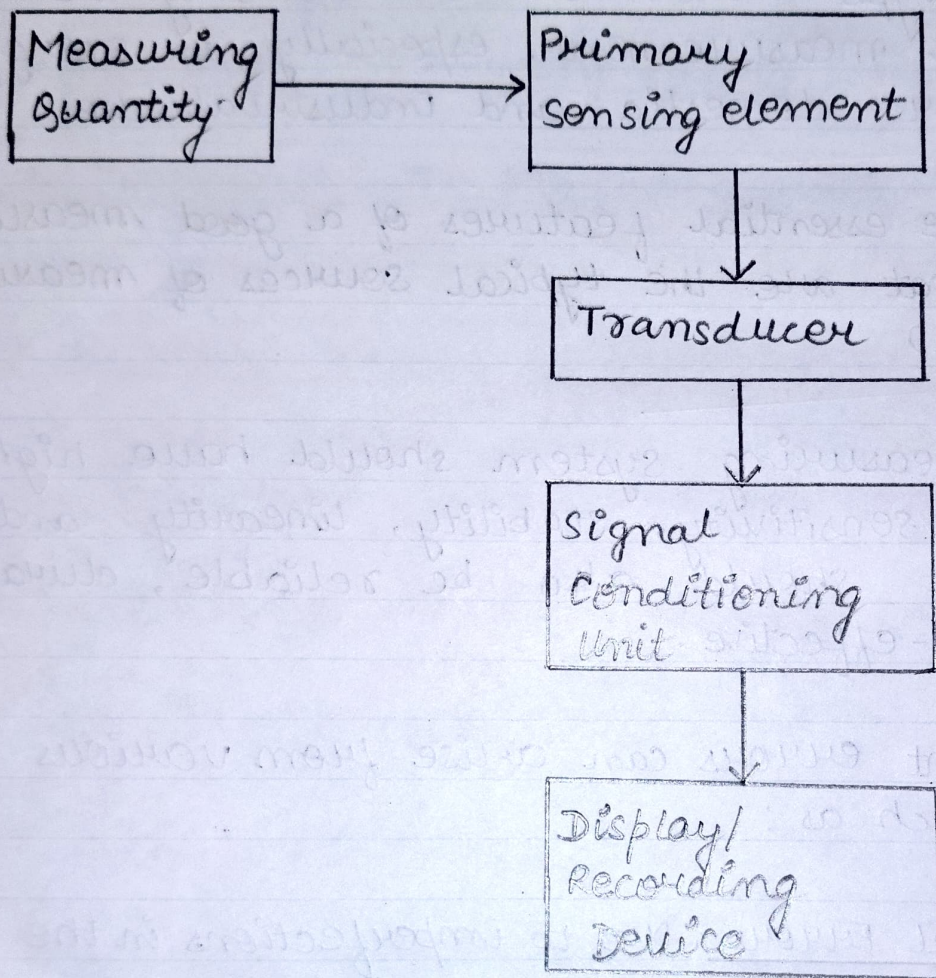
Induction-type wattmeter is commonly used in AC power measurement, especially in energy meters for domestic and industrial use.

8. Describe the essential features of a good measuring system. What are the typical sources of measurement errors?

Ans → A good measuring system should have high accuracy, sensitivity, stability, linearity and fast response. It should also be reliable, durable, and cost-effective.

Measurement errors can arise from various sources such as:

1. Instrumental Errors - Due to imperfections in the instrument
2. Environmental Errors - Caused by external factors like temperature, humidity.
3. Observational Errors - Human errors in reading or interpreting measurements.
4. Transmission Errors - Losses or distortions when signal travels through circuits or systems.



BLOCK DIAGRAM OF A BASIC MEASURING SYSTEM

9. With a neat diagram, explain the block diagram of a basic measuring system. Highlight the role of each element.

Ans → A basic measuring system consists of:

1. Measurand: The physical quantity to be measured.
(Eg: temperature, pressure, displacement).
2. Primary Sensing Element: Detects the measurand and gives a proportional response. (Eg: A bulb thermometer sensing temperature).
3. Transducer: Converts the sensed physical quantity into an electrical signal.
(Eg: Thermocouple (temperature $\rightarrow$ voltage)).
4. Signal Conditioning Unit: Modifies the signal to a suitable form for display - includes amplification, filtering and analog-to-digital conversion.
(Eg: Amplifier boosting thermocouple voltage).
5. Display / Recording Device: Shows or stores the final measured value.
(Eg: Digital display, data logger, or meter).
10. Define transducers and sensors. Explain the classification of transducers and their selection criteria in instrumentation.

Ans → Sensor → A sensor is a device that detects a physical quantity (like temperature, pressure or motion) and responds with a measurable signal.

Transducer → A transducer converts one form of energy into another, it typically converts a physical quantity into an electrical signal.

Classification of Transducers →

(a) Based on Energy source:

- Active Transducers: Generate output without external power.

Eg: Thermocouple (temperature → voltage).

- Passive Transducers: Require external power to operate.

Eg: Strain gauge (deformation → resistance change).

(b) Based on output signal:

- Analog Transducers: Give continuous output.

Eg: LVDT

- Digital Transducers: Give discrete / digital output.

Eg: Optical encoder.

(c) Based on Measurand:

- Temperature : Thermistor, RTD.
- Pressure : Piezoelectric sensor
- Displacement : LVDT
- Force : Load Cell

Their selection criteria include: range, sensitivity, linearity, repeatability, response time, stability and environmental compatibility. The right choice ensures precise, reliable, and durable performance in the intended measurement setup.